

Literature Status Note: Open Problem A3 in arXiv:2001.02226

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Status

literature_already_answered. This packet records an exact later literature answer, not a new proof produced by the run.

Source Question

The source is Petru Mironescu and Jean Van Schaftingen, “Trace theory for Sobolev mappings into a manifold”, arXiv:2001.02226, published in *Ann. Fac. Sci. Toulouse Math. (6)* 30 (2021), 281–299. In the introduction, Open Problem A3 asks whether the local trace space

$$W^{1-1/p,p}(B^{m-1}, N)$$

has the extension property when $4 \leq p < m$,

$$\pi_1(N), \dots, \pi_{\lfloor p-2 \rfloor}(N) \quad \text{are finite,} \quad \pi_{\lfloor p-1 \rfloor}(N) = 0.$$

The paper immediately notes that Bethuel–Demengel had conjectured a positive answer.

Answering Source

The supporting paper is Jean Van Schaftingen, “The extension of traces for Sobolev mappings between manifolds”, arXiv:2403.18738. Its introduction says that the paper gives a complete answer to trace surjectivity and proves that there is no further obstruction beyond the known topological and analytical ones. The same paper cites the 2021 Mironescu–Van Schaftingen paper.

The decisive local theorem is Theorem 1.1, labelled **theorem_extension_halfspace**. For $1 < p < m$, it states that

$$\mathrm{tr}_{\mathbb{R}^{m-1}}(\dot{W}^{1,p}(\mathbb{R}_+^m, N)) = \dot{W}^{1-1/p,p}(\mathbb{R}^{m-1}, N)$$

if and only if

$$\pi_1(N), \dots, \pi_{\lfloor p-2 \rfloor}(N) \quad \text{are finite,} \quad \pi_{\lfloor p-1 \rfloor}(N) = 0.$$

Moreover the theorem supplies a constant C so that the extension can be chosen with

$$\int_{\mathbb{R}_+^m} |DU|^p \leq C \iint_{\mathbb{R}^{m-1} \times \mathbb{R}^{m-1}} \frac{d(u(y), u(z))^p}{|y - z|^{p+m-2}} dy dz.$$

The collar formulation, Theorem 1.1’, gives the corresponding local manifold version.

Identification

Open Problem A3 is exactly the subrange $4 \leq p < m$ of the local surjectivity theorem above. Its homotopy hypotheses match the necessary and sufficient hypotheses in the 2024 theorem. Therefore A3 has a positive literature answer. The adjacent quantitative question in the source paper is also answered positively in the half-space/collar setting by the displayed linear estimate.

The source paper formulates the question on B^{m-1} after explaining that trace theory has local versions in which $\mathbb{R}^{m-1}$ is replaced by a Lipschitz domain. The 2024 half-space and collar theorems are the standard local model for that same trace-extension problem.

Scope and Limitations

This packet does not claim a new solution. It records that the target should be skipped by future agents as already answered in later literature. The global compact-manifold theorem in arXiv:2403.18738 has an additional skeleton-extension condition involving M and ∂M ; this note only identifies the local A3/linear-estimate answer.

References

- [1] P. Mironescu and J. Van Schaftingen, *Trace theory for Sobolev mappings into a manifold*, Ann. Fac. Sci. Toulouse Math. (6) 30 (2021), 281–299. arXiv:2001.02226.
- [2] J. Van Schaftingen, *The extension of traces for Sobolev mappings between manifolds*, arXiv:2403.18738.